The system’s temperature, in turn, is directly connected to the energy of the simulated system and the probability that it is allowed to transition to a higher-energy state (in other words, to a solution for which the cost function takes higher values).

Figure 17.4 shows a possible path on a 3D surface that resembles the progress of simulated annealing optimization: this time, the algorithm is capable of getting itself out of a local minimum, even by taking a counter intuiive step uphill: that’s the difference with greedy algorithms (including the ones summarized in section 17.1), and the true advantage of this technique.

The path shown in figure 17.4, however, though possible[6], it’s not typical for simulated annealing; a more common path would be much more chaotic – and would have made the figure a bit too messy.